SLIC Index V3

Methodology Technical Paper

English master edition

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This paper documents the live published SLIC scorer as implemented in **scripts/rescore-all-cities.mjs** and written to **src/data/publishedRankingData.json**. It replaces earlier draft descriptions of a 35-indicator anchor model that no longer matches the public dataset.

Snapshot fact	Value
Cities in public file	163
Ranked cities	154
Watchlist cities	9
Scored metric lines	20
Visible non-scoring diagnostics	3

1. Published method

SLIC has five public pillars. Within each pillar, observed scored metrics are combined by weighted mean. Across pillars, the final score uses a weighted AMPI, so cross-pillar imbalance is punished mathematically rather than rhetorically.

```
normalize(x, p05, p95, positive) = 100 * clamp((winsor(x) - p05) / (p95 - p05), 0, 1)
normalize(x, p05, p95, negative) = 100 * clamp((p95 - winsor(x)) / (p95 - p05), 0, 1)

P_p(c) = sum observed alpha_(p,m) * q_m(c) / sum observed alpha_(p,m)

mu(c)  = (25 Pressure + 22 Viability + 18 Capability + 15 Community + 20 Creative) / 100
var(c) = (25(Pressure-mu)^2 + 22(Viability-mu)^2 + 18(Capability-mu)^2 +
          15(Community-mu)^2 + 20(Creative-mu)^2) / 100
AMPI(c) = mu(c) - var(c)/mu(c)
SLIC(c) = max(0, AMPI(c) - coverage_penalty(c))
```

The final published score is therefore **not** a simple weighted average of the displayed pillar scores. That distinction matters: a city with one catastrophic pillar is supposed to score worse than a city with the same weighted mean but more even performance.

2. What is scored

The current public model scores **20** metric lines and keeps **3** additional metric lines visible as diagnostics only. Visibility and scoring are intentionally separated.

Growth (25)

Metric	Weight	Status	Method note
Tax-adjusted PPP disposable income	9	Scored	Residual money after essentials, converted once through the PPP private-consumption factor.
Housing burden	5	Scored	Housing cost share of income; higher burden scores worse.
Household debt burden	4	Scored	Debt relative to income, with fallback when city debt evidence is thin.
Working time pressure	4	Scored	Higher work burden scores worse.
Suicide and severe mental strain	3	Scored	Public-health strain proxy; higher severe-strain burden scores worse.
Economic growth momentum	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Viability (22)

Metric	Weight	Status	Method note
Personal safety	5	Scored	Harm and victimization outcomes, not surveillance theatre.
Transit access and commute burden	5	Scored	Better access and lower burden score better.
Clean air	4	Scored	Higher pollution exposure scores worse.
Water, sanitation, and utility reliability	4	Scored	Safer access and higher reliability score better.
Digital infrastructure	4	Scored	Broadband quality and affordability.
Climate and sunlight livability	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Capability (18)

Metric	Weight	Status	Method note
Healthcare quality	8	Scored	Effective-care quality and access.
Education quality	6	Scored	Learning quality and skills formation.

Metric	Weight	Status	Method note
Equal opportunity and distributional fairness	4	Scored	Composite scored metric: 70% equal-opportunity evidence, 30% reversed Gini context.

Community (15)

Metric	Weight	Status	Method note
Hospitality and belonging	5	Scored	Do people feel welcome in practice?
Tolerance and pluralism	5	Scored	Composite scored metric: 40% Equaldex, 30% Freedom House, 30% reversed hate-crime or civil-liberties proxy.
Cultural, historic, and public-life vitality	5	Scored	Everyday public-life texture; visitor demand is contextual, not an automatic bonus.
Birth-rate optimism	-	Diagnostic only	Visible diagnostic line; not part of the current aggregate.

Creative (20)

Metric	Weight	Status	Method note
Entrepreneurial dynamism	6	Scored	Startup formation and productive entry.
Innovation and research intensity	5	Scored	Research depth and knowledge production.
Economic vitality and productive context	5	Scored	Composite scored metric: 50% investment signal, 30% GDP per capita PPP, 20% GDP growth.
Administrative and investment friction	4	Scored	Higher time-cost and licensing burden score worse.

3. Composite metrics

Three scored terms are composites. When one component is missing, the composite is reweighted over the observed component weights only; nothing is silently imputed.

```
EqualOpportunityFairness = 0.7 * EqualOpportunity + 0.3 * ReverseGini
TolerancePluralism       = 0.4 * Equaldex + 0.3 * FreedomHouse + 0.3 * ReverseHateCrime
EconomicVitality         = 0.5 * InvestmentSignal + 0.3 * GDPperCapitaPPP + 0.2 * GDPGrowth
```

4. Coverage, watchlist logic, and ranking protocol

```
cov_direct = 1 if observed else 0
cov_composite = observed component weight / total component weight

Cov_p(c) = sum alpha_(p,m) * cov_m(c) / sum alpha_(p,m)
Cov(c)   = (25 Cov_pressure + 22 Cov_viability + 18 Cov_capability +
            15 Cov_community + 20 Cov_creative) / 100
```

Coverage grade	Rule	Penalty	Rank status
A	Cov >= 0.75	0	Ranked
B	0.50 <= Cov < 0.75	5	Ranked
C	0.35 <= Cov < 0.50	15	Ranked
Watchlist	Cov < 0.35 or manual watchlist override	0	Watchlist

After score computation, ranked cities receive a pure score rank: descending SLIC, then deterministic tie-breaks on Community, Pressure, and city label. Alpha, Beta, and Gamma are then computed as a separate public overlay. The overlay allows only one city per country across all three tiers; Alpha requires Community >= 40 and Pressure >= 40, with Europe capped at one Alpha seat and East Asia capped at one city from Japan and one from Korea; Beta raises the floors to Community >= 45 and Pressure >= 45; Gamma fills from the remaining ranked cities.

Live example: Singapore is currently pure rank **#18** with SLIC **60.2**, but its Community (38.8) and Pressure (43.3) place it in **Gamma**. Bangkok is pure rank **#26** with SLIC **58.5**, yet enters **Alpha** because Community (90.0) and Pressure (45.4) both clear the Alpha floor.

5. Worked example from the live dataset

Example city: **Kaohsiung**. This walkthrough uses the published row in the live dataset, not a fictional demonstration.

```
Negative-direction normalization example (personal safety)
raw = 0.8
p05 = 0.23
p95 = 22.63
score = 100 * (22.63 - 0.8) / (22.63 - 0.23)
score = 97.5

Growth weighted mean
Growth = (9x81.5 + 5x81.1 + 4x41.4 + 3x27.5) / 21
Growth = 66.1

Cross-pillar AMPI
mu  = 70.308
var = 252.971
AMPI = 70.308 - 252.971 / 70.308
AMPI = 66.710
Coverage = 0.77 -> grade A -> penalty 0
Published SLIC = 66.7
```

The important methodological point is visible in the arithmetic: the weighted pillar mean for this city is higher than the final score because AMPI makes uneven pillar structure costly.

6. Data profile of the current public file

Measure	Value
City / metro metric share	48.7%
National / international metric share	25.9%
Composite metric share	15.7%
Derived metric share	9.6%
Grade A cities	138
Grade B cities	2
Grade C cities	16
Watchlist cities	9

These shares are computed directly from non-null metric lines in the published JSON. They are included here so the paper does not rely on invented methodology graphics or unsupported source-mix claims.

7. Published ranked cities

The table below is the ranked slice of the current published dataset. Watchlist cities are excluded from this list by definition.

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
1	Raleigh	United States	69.0	75.1	76.0	79.1	53.8	62.0	A
2	Montreal	Canada	68.9	62.3	93.3	75.3	74.1	54.1	A
3	Kaohsiung	Taiwan	66.7	66.1	86.5	90.6	47.1	56.9	A
4	Minneapolis	United States	64.6	71.6	69.1	79.1	44.2	62.0	A
5	Pittsburgh	United States	64.5	76.0	65.1	79.1	50.2	55.7	A
6	Eindhoven	Netherlands	64.3	64.6	97.6	71.8	65.7	42.8	A
7	Graz	Austria	63.9	62.4	87.1	74.4	73.4	41.4	A
8	Taipei	Taiwan	63.8	48.4	89.5	89.7	47.1	69.4	A
9	Prague	Czechia	63.5	62.0	90.6	61.9	87.7	42.4	A
10	Ottawa	Canada	63.5	55.3	93.3	75.3	75.6	44.2	A
11	Jeju City	South Korea	61.6	56.2	99.2	85.8	44.8	50.6	A
12	Lyon	France	60.6	52.4	97.9	63.9	73.1	45.1	A
13	Braga	Portugal	60.6	58.4	95.7	68.8	57.9	43.8	A
14	Milan	Italy	60.5	88.5	85.8	66.9	58.4	29.4	A
15	Incheon	South Korea	60.5	53.3	99.2	85.8	44.8	50.6	A
16	Busan	South Korea	60.5	53.2	99.2	85.8	44.8	50.6	A
17	Suwon	South Korea	60.3	53.3	93.1	85.8	45.1	50.6	A
18	Singapore	Singapore	60.2	43.3	87.4	94.1	38.8	73.3	A
19	Copenhagen	Denmark	59.9	33.1	97.9	71.3	89.6	61.5	A
20	Perth	Australia	59.8	45.3	89.5	87.1	56.3	51.2	A
21	Tokyo	Japan	59.5	46.2	94.2	76.8	70.2	44.2	A
22	Toronto	Canada	59.3	38.7	93.3	75.3	72.5	54.1	A
23	Valparaiso	Chile	58.9	58.3	72.7	73.6	56.9	43.0	A
24	Porto	Portugal	58.6	52.5	95.7	68.8	57.9	43.8	A
25	Chicago	United States	58.6	56.0	56.6	79.1	47.6	62.0	A
26	Bangkok	Thailand	58.5	45.4	84.9	62.5	90.0	46.0	A
27	Adelaide	Australia	58.5	43.7	89.5	87.1	56.3	49.2	A
28	Zurich	Switzerland	58.3	39.1	97.6	72.7	76.0	49.4	A
29	London	United Kingdom	58.2	35.1	82.8	78.4	70.2	59.3	A
30	Helsinki	Finland	58.2	40.9	91.5	81.0	57.5	52.9	A
31	Brisbane	Australia	58.2	39.7	89.5	87.1	56.3	54.4	A
32	Melbourne	Australia	58.2	36.4	89.5	87.1	56.3	60.5	A
33	Tallinn	Estonia	58.2	51.0	90.1	58.4	49.0	57.8	A
34	Vienna	Austria	58.1	46.8	87.1	74.4	71.1	41.4	A
35	Santiago	Chile	57.9	50.7	72.7	83.8	56.9	44.6	A
36	New York	United States	57.7	27.1	85.4	82.0	68.5	75.3	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
37	Fukuoka	Japan	57.6	59.0	94.5	67.4	46.6	42.3	A
38	Dunedin	New Zealand	57.3	51.2	92.6	85.7	57.2	36.5	A
39	Vancouver	Canada	57.1	34.7	93.3	75.3	70.2	54.1	A
40	Christchurch	New Zealand	57.0	45.8	92.6	85.7	57.2	41.4	A
41	Venice	Italy	56.8	53.9	84.7	67.8	68.6	34.8	A
42	Torun	Poland	56.0	68.4	79.5	65.1	52.0	31.9	A
43	Wellington	New Zealand	55.9	37.8	92.6	85.7	57.2	48.8	A
44	Abu Dhabi	United Arab Emirates	55.9	70.5	92.6	62.0	29.5	46.2	A
45	Hiroshima	Japan	55.8	59.0	94.5	67.4	46.6	37.3	A
46	Sapporo	Japan	55.8	59.0	94.5	67.4	46.6	37.3	A
47	Dubai	United Arab Emirates	55.8	69.8	92.6	62.0	29.5	46.2	A
48	Katowice	Poland	55.4	65.9	79.5	65.1	52.0	31.9	A
49	Kobe	Japan	55.4	58.0	94.5	67.4	46.6	37.3	A
50	Manama	Bahrain	55.3	66.4	79.7	57.8	39.7	42.5	A
51	Sydney	Australia	54.5	27.8	89.5	87.1	56.3	63.9	A
52	Auckland	New Zealand	54.4	32.4	92.6	85.7	57.2	52.9	A
53	Gdansk	Poland	54.3	60.9	79.5	65.1	52.0	31.9	A
54	Bologna	Italy	54.0	56.4	88.2	66.9	58.4	29.4	A
55	Amsterdam	Netherlands	53.9	37.6	97.6	71.8	65.7	42.8	A
56	Bratislava	Slovakia	53.9	57.3	89.7	49.7	63.2	35.3	A
57	Hobart	Australia	52.5	33.8	89.5	87.1	56.3	45.1	A
58	Krakow	Poland	52.3	53.5	79.5	65.1	52.0	31.9	A
59	Port Louis	Mauritius	51.2	56.0	76.2	47.0	39.6	45.4	A
60	Bucharest	Romania	49.5	57.5	81.4	50.6	40.7	34.8	A
61	Melaka	Malaysia	49.5	63.0	75.3	52.4	40.3	31.5	A
62	Kuching	Malaysia	49.3	61.5	73.7	52.4	41.1	31.5	A
63	Kota Kinabalu	Malaysia	49.3	61.5	77.9	52.4	40.0	31.5	A
64	Montevideo	Uruguay	49.1	52.4	59.5	74.6	64.6	24.4	A
65	Khobar	Saudi Arabia	48.9	91.3	96.3	78.9	17.2	24.1	A
66	Jeddah	Saudi Arabia	48.8	90.6	96.3	78.9	17.2	24.1	A
67	Riyadh	Saudi Arabia	48.7	89.9	96.3	78.9	17.2	24.1	A
68	Paris	France	48.6	27.7	97.9	63.9	69.2	45.1	A
69	George Town	Malaysia	48.6	58.0	73.7	52.4	41.1	31.5	A
70	San Juan	Puerto Rico	48.2	43.8	59.2	68.3	63.3	45.8	B
71	Budapest	Hungary	48.1	47.2	93.1	55.5	63.5	26.2	A
72	Kuala Lumpur	Malaysia	46.8	51.0	73.7	52.4	41.1	31.5	A
73	Hat Yai	Thailand	46.6	55.1	66.7	56.4	52.2	22.8	A
74	Salalah	Oman	45.4	73.9	71.1	42.0	31.2	30.6	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
75	San Jose	Costa Rica	45.3	45.7	43.5	66.0	47.9	35.9	A
76	Male	Maldives	45.0	54.6	72.1	39.8	35.8	35.4	A
77	Phuket	Thailand	44.9	48.0	68.0	56.4	52.2	22.8	A
78	Muscat	Oman	44.7	66.9	71.1	42.0	31.2	30.6	A
79	Doha	Qatar	44.7	59.0	73.9	54.4	18.6	37.6	A
80	Cordoba	Argentina	44.5	67.5	69.2	78.6	46.8	10.2	A
81	Panama City	Panama	44.2	61.0	44.0	64.2	41.8	26.9	A
82	Chiang Mai	Thailand	43.8	54.3	49.7	56.4	52.2	22.8	A
83	Belgrade	Serbia	43.1	46.5	74.8	53.2	32.1	30.3	A
84	Tianjin	China	42.4	43.5	98.6	71.9	19.7	37.1	A
85	Curitiba	Brazil	42.2	51.8	45.1	69.5	36.3	27.6	A
86	Kuwait City	Kuwait	42.2	91.4	69.2	65.0	19.7	19.9	A
87	Florianopolis	Brazil	41.9	50.4	45.1	69.5	36.3	27.6	A
88	Buenos Aires	Argentina	41.8	55.4	69.2	78.6	46.8	10.2	A
89	Chongqing	China	41.0	38.5	75.2	71.9	20.3	37.1	A
90	Chengdu	China	40.9	37.8	77.0	71.9	20.5	37.1	A
91	Guangzhou	China	40.0	38.1	98.6	71.9	19.7	37.1	A
92	Sao Paulo	Brazil	39.9	42.7	45.1	69.5	36.3	27.6	A
93	Hangzhou	China	39.0	33.5	84.0	71.9	20.6	37.1	A
94	Shenzhen	China	38.7	35.3	98.6	71.9	19.7	37.1	A
95	Nanjing	China	38.4	52.6	75.3	59.0	7.1	34.5	A
96	Vilnius	Lithuania	36.9	65.0	-	41.6	74.0	41.4	C
97	Shanghai	China	36.7	31.3	98.6	71.9	19.7	37.1	A
98	Arequipa	Peru	36.3	57.5	34.1	42.3	34.9	24.7	A
99	Lima	Peru	35.8	51.3	34.1	42.3	34.9	24.7	A
100	Gaborone	Botswana	35.7	51.4	31.9	22.2	45.2	39.0	A
101	Suva	Fiji	35.7	45.8	36.1	47.7	30.8	25.1	A
102	Nizhny Novgorod	Russia	35.7	70.7	62.4	59.2	15.9	15.5	A
103	Medellin	Colombia	35.0	54.4	28.0	66.3	41.9	19.6	A
104	Bogota	Colombia	34.1	48.7	28.0	66.3	41.9	19.6	A
105	Cebu City	Philippines	33.2	55.8	38.2	40.4	30.8	17.5	A
106	Makati	Philippines	33.0	52.1	38.2	40.4	30.8	17.5	A
107	Windhoek	Namibia	32.9	42.7	30.5	32.1	39.3	25.4	A
108	Surabaya	Indonesia	32.7	63.7	40.8	35.1	29.9	17.8	A
109	Jakarta	Indonesia	32.7	58.6	40.8	35.1	29.9	17.8	A
110	Riga	Latvia	32.4	60.6	-	33.5	76.2	40.3	C
111	Rabat	Morocco	32.4	46.9	59.9	34.1	24.2	18.8	A
112	Casablanca	Morocco	32.3	46.0	59.9	34.1	24.2	18.8	A
113	Gothenburg	Sweden	31.5	65.3	-	14.5	95.5	63.3	C
114	Hyderabad	India	31.3	44.7	53.1	30.4	33.5	14.6	A

#	City	Country	SLIC	G	V	Cap	Com	Cr	Grade
115	Pune	India	31.2	44.4	53.1	30.4	33.5	14.6	A
116	Bengaluru	India	31.1	43.9	53.1	30.4	33.5	14.6	A
117	Chandigarh	India	30.6	43.9	44.1	30.4	34.1	14.6	A
118	Kigali	Rwanda	30.5	56.0	47.2	15.8	30.5	24.8	A
119	Moscow	Russia	30.5	41.0	62.4	59.2	15.9	15.5	A
120	Kandy	Sri Lanka	29.0	51.5	40.7	33.7	28.6	11.7	A
121	Colombo	Sri Lanka	28.9	48.8	40.7	33.7	28.6	11.7	A
122	Aqaba	Jordan	28.9	43.2	66.0	46.5	11.5	17.0	A
123	Thimphu	Bhutan	28.6	47.2	47.0	25.2	40.4	10.2	A
124	Da Nang	Vietnam	28.2	51.7	-	44.3	28.1	20.8	B
125	Amman	Jordan	28.1	40.3	66.0	46.5	11.5	17.0	A
126	Tbilisi	Georgia	26.7	50.0	-	33.1	40.9	43.1	C
127	Valencia	Spain	26.6	59.8	-	31.1	87.1	30.4	C
128	Johannesburg	South Africa	26.3	34.3	15.1	26.3	42.3	29.8	A
129	Cape Town	South Africa	26.2	34.2	15.1	26.3	42.3	29.8	A
130	Zagreb	Croatia	25.7	61.9	-	17.8	77.4	42.3	C
131	Accra	Ghana	25.3	55.6	35.9	14.9	30.3	17.8	A
132	Merida	Mexico	25.1	53.7	13.3	55.5	40.5	14.1	A
133	Kumasi	Ghana	25.1	59.4	35.9	14.9	30.3	17.8	A
134	Guadalajara	Mexico	24.8	49.9	13.3	55.5	40.5	14.1	A
135	Cork	Ireland	24.5	82.1	-	13.3	91.8	37.4	C
136	Nairobi	Kenya	23.9	48.5	45.1	12.5	28.1	14.5	A
137	Mexico City	Mexico	23.3	39.2	13.3	55.5	40.5	14.1	A
138	Pokhara	Nepal	22.3	44.3	37.6	12.2	21.7	16.6	A
139	Kathmandu	Nepal	22.3	42.3	37.6	12.2	21.7	16.6	A
140	Bergen	Norway	21.7	76.4	-	3.2	97.8	48.8	C
141	Chattogram	Bangladesh	21.7	51.3	48.3	10.8	19.6	15.7	A
142	Dhaka	Bangladesh	21.6	49.3	48.3	10.8	19.6	15.7	A
143	Asuncion	Paraguay	17.5	40.4	-	74.7	45.4	14.9	C
144	Antwerp	Belgium	16.9	65.1	-	4.4	87.9	40.4	C
145	Phnom Penh	Cambodia	16.6	43.9	-	18.2	34.5	36.3	C
146	Brno	Czechia	15.5	70.5	-	0.0	84.3	42.4	C
147	Dakar	Senegal	11.4	40.5	0.7	10.6	26.3	26.7	A
148	Munich	Germany	10.8	18.3	-	27.1	65.0	36.2	C
149	Islamabad	Pakistan	10.7	47.5	32.7	8.1	13.0	4.4	A
150	Lahore	Pakistan	10.6	48.3	32.7	8.1	13.0	4.4	A
151	Karachi	Pakistan	10.6	48.0	32.7	8.1	13.0	4.4	A
152	Ljubljana	Slovenia	10.2	67.0	-	0.0	90.6	33.2	C
153	Kampala	Uganda	8.7	31.8	-	68.7	15.1	22.5	C
154	Alexandria	Egypt	2.4	45.0	-	11.3	7.7	30.6	C